



Lab Report

Only for course Teacher						
		Needs Improvement	Developing	Sufficient	Above Average	Total Mark
Allocate mark & Percentage		25%	50%	75%	100%	5
Level of Content	1					
Development	2					
Spelling & Grammar	1					
Organization and Formatting	1					
Total obtained mark						
Comments						

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Course Code: SE 334

Course Name: Artificial Intelligence Lab

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PROJECT SYNOPSIS

Project Title:
**NutriNLP: An Intelligent Food Calorie
Prediction and Nutrition
Recommendation System Using Natural
Language Processing**

Expected Contribution:

Payel Saha ID: 232-35-275	Data Cleaning & Preprocessing	Model Development 1. Logistic Regression 2. Multinomial Naive Bayes	Model evaluation	Testing
Mst Sayida Parvin ID: 232-35-280	NLP Preprocessing (Tokenization, Lemmatization)	Model Development 1. Decision Tree 2. Random Forest 3. Support Vector Machine (SVM)	Model evaluation	Testing
Afifa Rahman ID: 232-35-544	NLP Preprocessing (TF- IDF)	Model Development 1. K-Nearest Neighbors (KNN) 2.Gradient Boosting 3. Extreme Gradient Boosting (XGBoost)	Model evaluation	Testing

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1. Introduction

Healthy eating has become increasingly important, yet many people lack sufficient nutritional knowledge to make informed dietary decisions. While several nutrition applications exist, most require users to manually search for food items or depend on predefined databases without understanding natural language descriptions.

Natural Language Processing (NLP) enables computers to understand human language and extract meaningful information from textual input. This project proposes an intelligent food recommendation system that accepts food descriptions written in natural language, predicts the calorie category and nutritional information of the meal, and provides healthier dietary recommendations.

Using the Nutrition5K dataset, ingredient lists are transformed into textual representations suitable for NLP processing. The system applies text preprocessing, feature extraction using TF-IDF, and multiple machine learning classification algorithms to predict nutritional characteristics and recommend healthier meal alternatives.

The project demonstrates how NLP can be applied to solve real-world healthcare and nutrition-related problems through intelligent text understanding and recommendation.

2. Motivation

Many individuals struggle to estimate the nutritional value of their meals, especially when preparing homemade dishes. Existing calorie tracking applications often require users to manually search for ingredients or rely on barcode scanning, making them less convenient for everyday use.

With the rapid advancement of Artificial Intelligence and NLP, it is now possible to interpret natural language food descriptions and automatically estimate nutritional information. This project aims to bridge the gap between nutrition awareness and AI by creating a system capable of understanding ingredient-based food descriptions and generating meaningful health recommendations.

Additionally, the project provides practical experience in applying NLP techniques, machine learning models, and recommendation systems to solve a real-world problem.

3. Objectives

The primary objective of this project is to develop an NLP-based intelligent nutrition recommendation system capable of understanding textual food descriptions and predicting calorie-related information.

Specific objectives include:

- Develop an NLP pipeline for processing ingredient-based food descriptions.
- Perform text preprocessing including tokenization, normalization, and stop-word removal.
- Convert textual ingredients into numerical feature vectors using TF-IDF.
- Train multiple machine learning models for calorie category prediction.
- Compare different classification algorithms based on performance metrics.
- Generate nutrition summaries including calories, protein, carbohydrates, and fat.
- Recommend healthier alternatives based on nutritional analysis.
- Build a user-friendly interface for food analysis and recommendation.

4. Problem Statement

Many people consume meals without understanding their nutritional composition, calorie intake, or overall health impact. Existing calorie estimation tools are often limited because they require manual food selection, image-based recognition or extensive user interaction.

Furthermore, most systems do not effectively process natural language descriptions such as:

"Chicken breast with rice and broccoli"

Without intelligent text understanding, users cannot quickly obtain nutritional insights or healthier recommendations.

Therefore, there is a need for an NLP-based intelligent system capable of analyzing ingredient-based text descriptions and automatically predicting nutritional information while suggesting healthier alternatives.

5. Proposed Solution

This project proposes NutriNLP, an AI-powered nutrition recommendation system that combines Natural Language Processing and Machine Learning to analyze food descriptions.

The system workflow includes:

- Accepting food descriptions from users.
- Performing NLP preprocessing.
- Extracting textual features using TF-IDF.
- Training multiple classification algorithms.
- Predicting calorie categories.
- Displaying estimated nutritional values.
- Generating healthier food recommendations.

Unlike conventional calorie calculators, the proposed system understands natural language ingredient descriptions and produces intelligent recommendations automatically.

6. Methodology / Workflow

The proposed NutriNLP system follows a structured Natural Language Processing (NLP) and Machine Learning workflow to analyze ingredient-based food descriptions, predict calorie categories, and generate nutrition recommendations. The overall methodology consists of the following phases:

Step 1: Data Collection

The project will utilize the Nutrition5K dataset as the primary data source. The dataset contains detailed information about food dishes, including ingredient lists and nutritional values such as

Step 2: Data Preprocessing

Raw textual data cannot be directly processed by machine learning algorithms. Therefore, several preprocessing techniques will be applied to improve data quality and prepare the text for feature extraction.

The preprocessing steps include:

1. Removing duplicate and missing records.
2. Combining all ingredients of a dish into a single sentence.
3. Converting all text to lowercase for consistency.
4. Removing punctuation marks and unnecessary symbols.
5. Tokenizing the ingredient sentences into individual words.
6. Removing stop words that do not contribute meaningful information.
7. Applying lemmatization to reduce words to their base forms.
8. Cleaning extra spaces and irrelevant characters.

Example:

Raw Input:

- Chicken, Rice, Broccoli, Butter

Processed Output:

- chicken rice broccoli butter

Step 3: Feature Engineering

After preprocessing, the textual ingredient descriptions will be transformed into numerical feature vectors using the Term Frequency–Inverse Document Frequency (TF-IDF) technique.

TF-IDF measures the importance of each ingredient within a dish while reducing the influence of commonly occurring ingredients. This numerical representation allows machine learning algorithms to analyze textual data effectively.

Each processed food description will therefore be represented as a high-dimensional feature vector suitable for model training.

Step 4: Model Development and Training

The transformed TF-IDF feature vectors will be divided into training and testing datasets. Multiple supervised machine learning algorithms will then be trained to classify each dish into one of three calorie categories:

1. Low Calorie
2. Medium Calorie
3. High Calorie

The following classification models will be implemented and compared:

1. Logistic Regression
2. Multinomial Naïve Bayes
3. Decision Tree
4. Random Forest
5. Support Vector Machine (SVM)
6. K-Nearest Neighbors (KNN)
7. Gradient Boosting
8. Extreme Gradient Boosting (XGBoost)

Training multiple models will allow the identification of the most accurate and reliable classifier for this task.

Step 5: Model Evaluation

Each trained model will be evaluated using the testing dataset to measure its predictive performance.

The evaluation metrics include:

1. **Accuracy:** Measures the overall correctness of predictions.
2. **Precision:** Evaluates how many predicted calorie categories are correct.
3. **Recall:** Measures the model's ability to correctly identify each calorie category.
4. **F1-Score:** Provides a balanced evaluation of precision and recall.
5. **Confusion Matrix:** Visualizes correct and incorrect classifications for each calorie category.

The model achieving the highest overall performance will be selected for deployment in the recommendation system.

Step 6: Nutrition Analysis and Recommendation Engine

After the best-performing model predicts the calorie category of a user-entered food description, the recommendation engine will generate meaningful nutritional insights.

The system will provide:

1. Estimated calorie category (Low, Medium, or High)
2. Nutritional summary including calories, protein, carbohydrates, and fat
3. Healthier ingredient substitutions (e.g., replacing white rice with brown rice)
4. Dietary improvement suggestions based on the nutritional composition
5. General nutrition recommendations to encourage healthier eating habits

This recommendation component enhances the practicality of the system by helping users make informed dietary decisions rather than simply displaying nutritional information.

7. Implementation Process

The project will be implemented in the following development phases:

Phase 1: Requirement Analysis and Project Planning: Define the project objectives, study existing nutrition recommendation systems, identify system requirements, and prepare the development plan.

Phase 2: Dataset Preparation and Exploratory Data Analysis (EDA): Collect the Nutrition5K dataset, examine its structure, analyze ingredient distributions, identify missing values, and understand nutritional attributes.

Phase 3: Data Preparation and NLP Pipeline Development: Clean the dataset, preprocess ingredient text, and construct the NLP pipeline for transforming raw ingredient descriptions into machine-readable features.

Phase 4: Machine Learning Model Development: Implement, train, and fine-tune multiple machine learning algorithms using TF-IDF features to predict calorie categories.

Phase 5: Recommendation Module Development: Develop a nutrition recommendation engine that generates nutritional summaries and suggests healthier ingredient substitutions based on prediction results.

Phase 6: User Interface Development: Design and implement a simple, interactive interface that allows users to enter food descriptions and view predictions and recommendations.

Phase 7: System Integration and Testing: Integrate all system components, conduct functional and performance testing, validate prediction results, and refine the system based on testing outcomes.

Phase 8: Documentation and Final Deployment: Prepare the project documentation, evaluate the overall system, and deploy the final application for demonstration.

8. Expected Result

The proposed system is expected to:

1. Correctly understand ingredient-based food descriptions.
2. Predict calorie categories with high accuracy.
3. Display nutritional information including calories, protein, carbohydrates, and fat.
4. Recommend healthier ingredient substitutions.
5. Compare the performance of multiple NLP-based machine learning models.
6. Demonstrate the practical application of NLP in nutrition analysis.

9. Conclusion

The proposed NutriNLP system demonstrates how Natural Language Processing can be effectively integrated with Machine Learning to develop an intelligent nutrition recommendation platform. By converting ingredient-based food descriptions into meaningful textual representations, the system can accurately predict calorie categories and provide valuable nutritional insights.

The project not only highlights the practical application of NLP in healthcare and dietary management but also offers a scalable foundation for future enhancements such as multilingual support, personalized meal planning, real-time nutrition tracking, and transformer-based language models. Ultimately, the system aims to promote healthier eating habits while showcasing the potential of AI-driven decision support in everyday life.